

Simulation [Frankly Speaking ...]

6-7 minutes

frank:datascience:simulation

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Prepare Sample Data

We will perform 1000 simulations of 40 observations from the exponential distribution with rate parameter (λ) of 0.2. For each of the simulations we will:

- calculate the samples means
- calculate the samples variances
- normalise the results for later comparison to the standard normal

```
set.seed(040)      # seed for random number generation
lambda <- 0.2      # exponential rate parameter
observ <- 40       # number of observations from exponential distribution
samples <- 1000    # number of samples of observations

# simulated data //1000// sample rows by //40// observation columns ...
simdata <- matrix(replicate(samples, rexp(observ, lambda)), samples, observ)

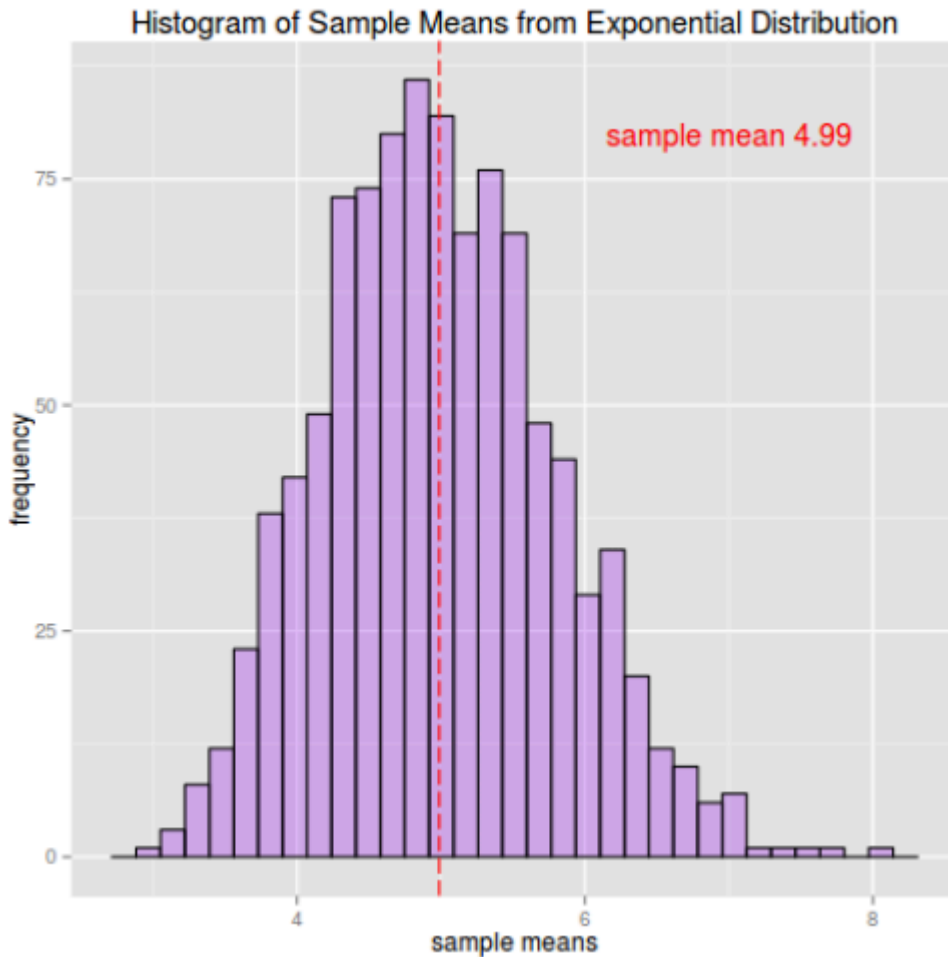
simmeans <- apply(simdata, 1, function(x) mean(x[])) # mean of each sample
simvars <- apply(simdata, 1, function(x) var(x[]))   # variance of each sample

# shift and scale data, i.e. normalise
mu = 1/lambda      # theoretical mean
se = sqrt(1/lambda ^ 2)/sqrt(observ)                # theoretical standard error
simnormals <- apply(simdata, 1, function(x) (mean(x[]) - mu)/se) # normalised samples

# put into data frame for use in plots
simdf <- data.frame(means = simmeans, vars = simvars, normals = simnormals)
```

Sample Mean versus Theoretical Mean

The method used was to make 40 observations of the exponential distribution repeating this 1000 times. Then take the mean for each sample of observations. The mean of these samples should approach the theoretical mean, $(1 / \lambda)$, as shown in the plot below:

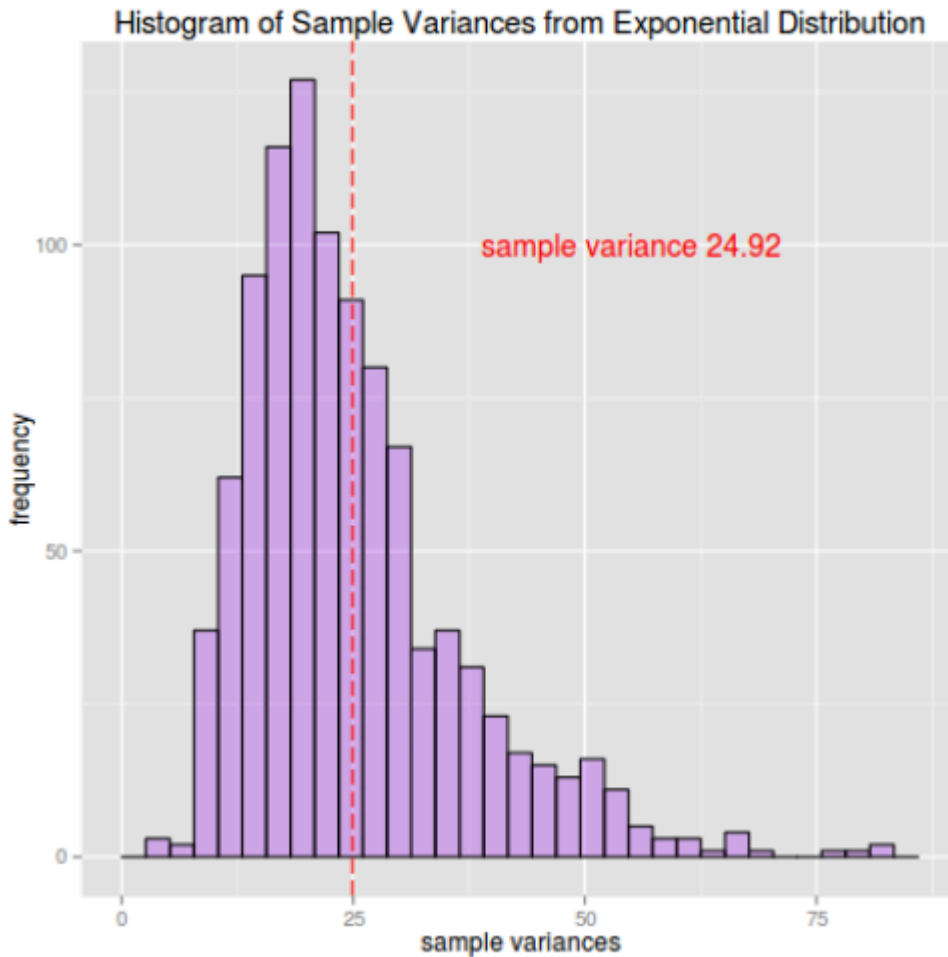


```
require(ggplot2, quietly = TRUE)
png(filename = "media/frank/datascience/simulation-sample-mean.png")
ggplot(simdf, aes(x=means)) +
  geom_histogram(colour = "black", fill = "purple", alpha = 0.3) +
  geom_vline(xintercept = mean(simdf$means), colour = "red", linetype = "longdash") +
  annotate("text", label = paste("sample mean", round(mean(simdf$means), 2)), x = 7, y = 80,
  colour = "red") +
  labs(x = "sample means", y = "frequency") +
  ggtitle("Histogram of Sample Means from Exponential Distribution")
dev.off()
```

- The exponential distribution has a theoretical mean of $1 / \lambda = 5$.
- The sample mean after 1000 samples is 4.99.

Sample Variance versus Theoretical Variance

As before, make 40 observations of the exponential distribution repeating this 1000 times. Then take the variance for each sample of observations. The mean of these sample variances should approach the theoretical variance, $(1 / \lambda^2)$ as shown in the plot below:

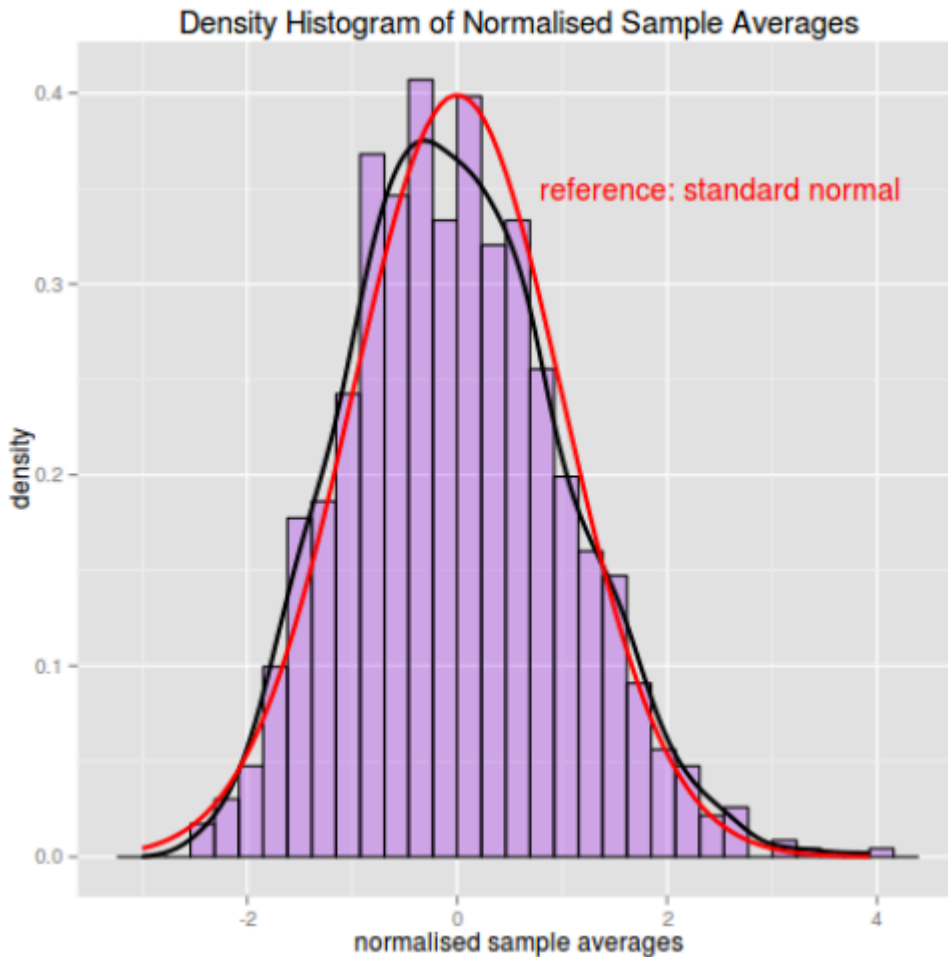


```
require(ggplot2, quietly = TRUE)
simvar <- mean(simdf$vars)
png(filename = "media/frank/datascience/simulation-sample-variance.png")
ggplot(simdf, aes(x=vars)) +
  geom_histogram(colour = "black", fill = "purple", alpha = 0.3) +
  geom_vline(xintercept = simvar, colour = "red", linetype = "longdash") +
  annotate("text", label = paste("sample variance", round(simvar, 2)), x = 55, y = 100,
  colour = "red") +
  labs(x = "sample variances", y = "frequency") +
  ggtitle("Histogram of Sample Variances from Exponential Distribution")
dev.off()
```

- The exponential distribution has a theoretical variance of $1 / \lambda^2 = 25$.
- The sample variance after 1000 samples is 24.92.

Distribution is Approximately Normal

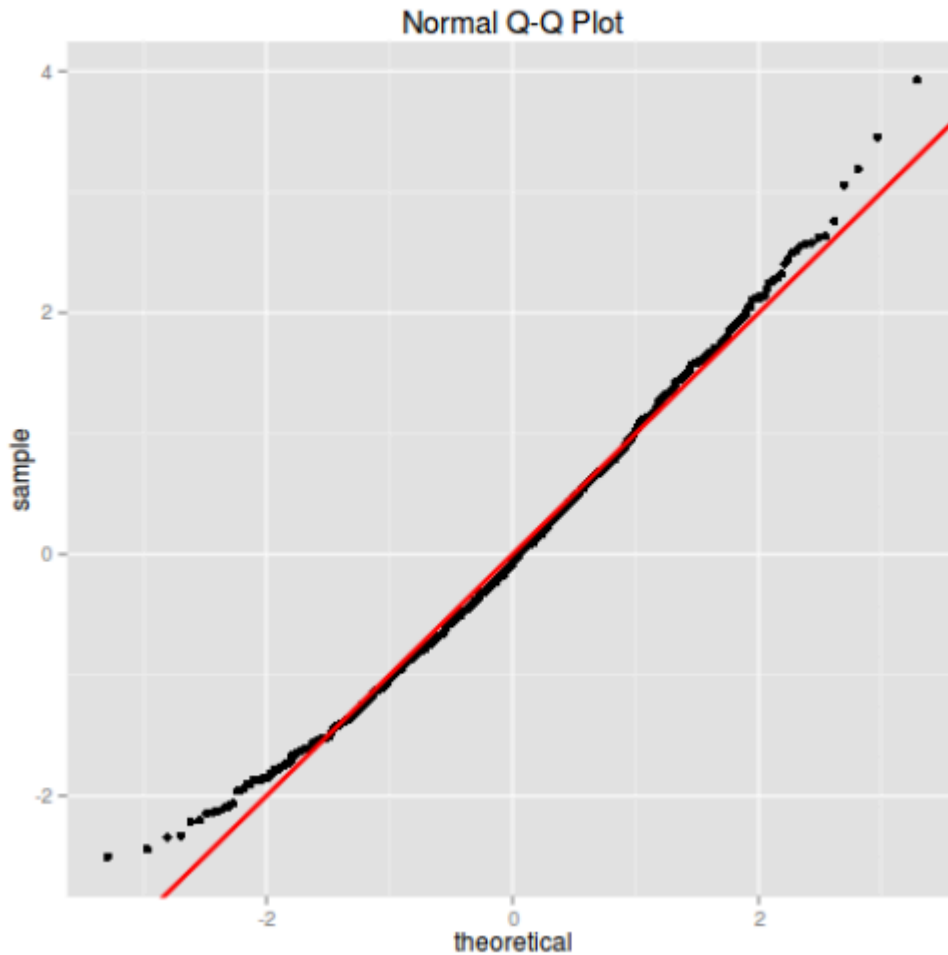
If we consider the normalised version, $\frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}}$, of our sample averages, then the Central Limit Theorem states that this distribution of averages tends to that of a standard normal as the sample size (n) increases. Plotting a density histogram of these averages we should see a normal like distribution:



```
require(ggplot2, quietly = TRUE)
png(filename = "media/frank/datascience/simulation-sample-normalised.png")
ggplot(data = simdf, aes(normals)) +
  geom_histogram(aes(y = ..density..), colour = "black", fill = "purple", alpha = 0.3) +
  geom_line(aes(y = ..density..), stat = "density", size = 1) +
  stat_function(fun = dnorm, colour = "red", size = 1, aes(x = c(-3, 3))) +
  labs(x = "normalised sample averages") +
  annotate("text", label = "reference: standard normal", x = 2.5, y = 0.35, colour = "red") +
  ggtitle("Density Histogram of Normalised Sample Averages")
dev.off()
```

To help evaluate how close this is to our simulated samples, we have overlaid the normalised sample averages with the standard normal distribution (in red), $N(0, 1)$.

We can further test this using the [Quantile-Quantile Plots](#). As you can see the sample [quantiles](#) (black dots) closely match the standard normal (red line):



```
require(ggplot2, quietly = TRUE)
png(filename = "media/frank/datascience/simulation-sample-quantiles.png")
qplot(sample = simdf$normals, stat = "qq") +
  geom_abline(intercept = 0, size = 1, colour = "red") +
  ggtitle("Normal Q-Q Plot")
dev.off()
```

compare with

```
# qqnorm(simdf$normals)
# qqline(simdf$normals, col = 2, lwd = 2)
```

Shapiro-Wilk normality test

The [Shapiro-Wilk test](#) tests the [null hypothesis](#) that a sample $x_1, \dots, x_n$ came from a normally distributed population.

For the example above we have

```
shapiro.test(simdf$normals)
```

Shapiro-Wilk normality test

```
data:  simdf$normals
W = 0.99299, p-value = 0.0001168
```

So, even though we have shifted and scaled the data, this does not make it normally distributed. Indeed the p-value is much less than 0.05, so we reject the null hypothesis that this is normally distributed.

Compare data that is normally distributed

```
shapiro.test(rnorm(100, mean = 5, sd = 3))
```

Shapiro-Wilk normality test

data: rnorm(100, mean = 5, sd = 3)

W = 0.9872, p-value = 0.4511

frank/datascience/simulation.txt

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